

Assignment#3

Problem Statement

Write a procedure Print that takes the address of a string and screen coordinates as parameters through the stack and displays the string on the screen at the specified coordinates. If the line ends, it prints the remaining text at the beginning of a new line. In the case of the last line, it scrolls the screen up by shifting all rows upward, moving line(n) to line(n-1) and discarding line(0). Also, call the print procedure from the main procedure to check its validity.

The C/C++ code for the print procedure is given below:

```

void print(char* string, int r, int c)
{
    int k=0;
    while(string[k] != '$')
    {
        screen[r][c] = string[k];
        k++;
        c++;
        if(c>79)
        {
            r++;
            if(r>24)
            {
                scroll();
                c=0; r=24;
            }
        }
    }
}

void scroll()
{
    int i,j;
    for(i=0;i<24;i++)
        for(j=0;j<80;j++)
            screen[i][j] = screen[i+1][j];
    for(j=0;j<80;j++)
        screen[24][j] = ' ';
}

```